Airport Data Analysis Using SQL

- ***Project Overview***

  This project analyses airport & airline data using sql to uncover patterns in airport performance, airline performance, route demand, passenger traffic across the airline network. The analysis focuses on high traffic routes, passenger flow between cities, passenger activity by airports, Airline performance across the routes. The goal is to generate actionable insights & recommendations that could help airlines improve route planning for better proficiency, optimise capacity allocation according to demand for efficiency as well as for cities & airports to plan their operations and marketing accordingly & infrastructure efficiently to improve their performance.

- **Dataset Description**

  The dataset contains airline operational data including passenger counts, freight, mail, distance, airline identifiers, origin and destination airports with their cities and states and time attributes such as year, quarter, and month.

- **Business Questions Addressed**

  - Analyze total passenger traffic per route and over time.

  - Determine average passengers per flight for various routes and airports

  - Assess flight frequency and identify high-traffic corridors.

  - Compare passenger numbers across origin cities to identify top-performing airports.

  - Identify popular destination airports based on inbound passenger counts.

- Rank routes which generate the most passenger traffic

- Which routes have low passenger volume but still operated frequently, underutilized routes

- Which airlines dominate passenger traffic, Airline market share analysis

- Which airports have highest passenger growth potential, Airports with high flight frequency but relatively low passengers

- Which Airport connect to the largest number of destinations, Network Connectivity Analysis

- What are Average passengers per flight by airline

- **Key SQL Queries & Results**

Top 10 busiest routes by passengers

ORIGIN_CITY_NAME	DEST_CITY_NAME	total_passengers
Miami, FL	New York, NY	156277
New York, NY	Miami, FL	149209
Orlando, FL	New York, NY	130145
New York, NY	Orlando, FL	125937
New York, NY	Chicago, IL	121251
New York, NY	Atlanta, GA	119729
Atlanta, GA	New York, NY	118629
Chicago, IL	New York, NY	116245
Orlando, FL	Atlanta, GA	108838
Fort Lauderdale, FL	New York, NY	108414

Top 5 airlines by market share

AIRLINE_ID	UNIQUE_CARRIER_NAME	total_passengers	rnk
19393	Southwest Airlines Co.	10998054	1
19790	Delta Air Lines Inc.	9739964	2
19805	American Airlines Inc.	9548187	3
19977	United Air Lines Inc.	8089034	4
20304	SkyWest Airlines Inc.	3196083	5

Top 10 origin airports by passenger volume

ORIGIN_AIRPORT_ID	ORIGIN_CITY_NAME	TOTAL_FLIGHTS	TOTAL_PASSENGERS
10397	Atlanta, GA	390	3095254
11292	Denver, CO	417	2728395
11298	Dallas/Fort Worth, TX	424	2575266
13930	Chicago, IL	571	2139178
13204	Orlando, FL	280	2001905
12889	Las Vegas, NV	347	1969324
14107	Phoenix, AZ	363	1811982
11057	Charlotte, NC	300	1793744
12892	Los Angeles, CA	345	1774710
14747	Seattle, WA	280	1502472

Underutilized routes (high frequency, low passengers)

route	origin_airport	dest_airport	total_flights	total_passengers	avg_passengers_per_flight
Anchorage, AK - Miami, FL	ANC	MIA	5	0	0
Fairbanks, AK - Rampart, AK	FAI	RMP	5	24	4.8
Newtok, AK - Bethel, AK	WWT	BET	5	82	16.4
Bethel, AK - Newtok, AK	BET	WWT	5	117	23.4
Toksook, AK - Bethel, AK	OOK	BET	5	118	23.6
Kongiganak, AK - Bethel, AK	KKH	BET	5	125	25
Tununak, AK - Bethel, AK	TNK	BET	5	125	25
Bethel, AK - Nunapitchuk, AK	BET	NUP	5	127	25.4
Bethel, AK - Akiachak, AK	BET	KKI	5	130	26
Bethel, AK - Kasigluk, AK	BET	KUK	5	143	28.6

Average passengers per flight by airline

AIRLINE_ID	UNIQUE_CARRIER_NAME	total_flights	avg_passengers
19805	American Airlines Inc.	868	11000.215437788018
19690	Hawaiian Airlines Inc.	94	8654.5
19790	Delta Air Lines Inc.	1265	7699.57628458498
20409	JetBlue Airways	328	6537.088414634146
20416	Spirit Air Lines	435	5823.255172413793
19977	United Air Lines Inc.	1394	5802.750358680058
19930	Alaska Airlines Inc.	480	4847.152083333333
19393	Southwest Airlines Co.	2470	4452.653441295546
20436	Frontier Airlines Inc.	628	3642.859872611465
20397	PSA Airlines Inc.	384	2716.7317708333335

Passenger growth potential airports (high flights, low passengers)

airport_id	city_name	total_flights	total_passenger_traffic	passengers_per_flight
13930	Chicago, IL	1138	4339064	3812.89
11298	Dallas/Fort Worth, TX	854	5165309	6048.37
11292	Denver, CO	830	5442961	6557.78
10397	Atlanta, GA	779	6221781	7986.88
14107	Phoenix, AZ	721	3632477	5038.11
12889	Las Vegas, NV	692	3881800	5609.54
12892	Los Angeles, CA	686	3574799	5211.08
12266	Houston, TX	610	2596313	4256.25
11057	Charlotte, NC	602	3601829	5983.1
13487	Minneapolis, MN	587	2183591	3719.92

- **SQL Techniques Used**

- Common Table Expressions (CTEs)

- Window Functions (ROW_NUMBER, RANK)

- Aggregations (SUM, AVG, COUNT)

- Joins (INNER JOIN, LEFT JOIN)

- UNION operations

- Data grouping and ranking

- **Key Insights from the Analysis**

- Certain routes carry significantly high passenger volumes compared to others, indicating concentrated demand of travelling between specific cities

- There are certain origin airports which are high performing comparatively with high numbers of passengers & flights being handled & departed from there

- We have certain high in demand destination cities & their airports with high number of passengers arriving there

- We have found certain routes between cities which are underutilised where they have low passenger volume but have flight frequency, they have got growth potential

- There are specific airlines which are dominating and performing well with having high passenger volumes with them compared to others

- There are airports which have huge connectivity with large number of destinations connected with it

- **Business Recommendations**

- **Increase capacity in high demand routes**

- Flight frequency or aircrafts with larger capacity must be deployed to those routes in order to capture the demand

- **Strengthen operations at major hub airport**

- Airports which are handling high passenger volumes should receive operational and infrastructure attention to maintain and uplift their service quality

- **Improving efficiency where there is underutilisation**

- Routes which are underutilised with low passenger volume but do have flight frequency, those locations must be advertised & marketed

properly to increase passenger volume as well as for Airlines, they must look after deploying smaller aircrafts and assign flights accordingly of demand to operate efficiently

- **Paying attention on airports having growth potential**

  - Marketing campaigns must be run as well as dynamic pricing can be addressed by lowering fares on low load flights to boost demand as well as necessary strategies can be made to increase growth on high potential areas by increasing passenger volume

- **Dominating airlines must run operations accordingly to maintain & improve their performance**

  - Airlines which are performing well with having them in high demand and having high passenger volume must plan strategies to maintain and improve their performance by increasing frequency of flight, strategize better deals & discounts on fares, if possible allocate larger aircrafts and their best crew for increasing their current capacity with proficiency.